Predictive Modeling

1 Introduction

Life expectancy is influenced by a range of economic, demographic, and social conditions. The objective of this exercise is to develop a predictive model for life expectancy using socioeconomic characteristics of different geographical areas.

The analysis involves cleaning and transforming the data, handling missing values and categorical variables, engineering relevant features, and comparing alternative predictive models. Model performance is evaluated using Mean Absolute Error (MAE) through cross-validation, with the final objective of generating life-expectancy predictions for the test dataset.

2 Data Overview

The dataset[1] contains information on factors such as land use, urban population, national income, inflation, internet and mobile access, sanitation, and representation of women in parliament. Each observation represents a geographical area, which may correspond to a country, region, or group of countries. The target variable is life expectancy.

- The training dataset contains 362 observations and 16 variables, with life expectancy as the target variable.
- Average life expectancy is 71.1 years, with values ranging from 48.9 to 83.5 years.
- Several variables, including surface area, agricultural land, forest area, armed forces, and secure internet servers, are highly right-skewed.
- Urban population varies substantially across geographical areas.
- Inflation variables contain substantial missing information, particularly weekly inflation.

2.1 Data Cleaning

The dataset contains a combination of numerical and categorical variables, with missing values and some variables reported in inconsistent formats. The following steps are used to prepare the data for modeling:

- Inflation: Annual, monthly, and weekly inflation contain substantial missing values. Missing annual inflation is recovered where possible by annualizing the available monthly or weekly inflation rates using compounding: $(1 + \pi_m)^{12} - 1$ for monthly inflation and $(1 + \pi_w)^{52} - 1$ for weekly inflation, where the inflation rates are expressed in decimal form. The monthly and weekly variables are then dropped.

[1] Data Source: HackerRank

- Categorical variables: The variables national income, number of mobile subscriptions, access to sanitation, and share of women in parliament have naturally ordered categories and are therefore converted to numerical values using ordinal encoding.
  - Unknown categories: For categorical variables containing "unknown" or "no info", a separate indicator variable is created to preserve information about whether the value was originally missing. The unknown category itself is then converted to 'NaN' for subsequent imputation.
- Internet usage: The number of internet users is reported using different denominators, such as users per 100 people and users per 1,000 people. These values are converted to a common proportion before modeling.
- Land variables: The variables surface area, agricultural land, and forest area are highly correlated. Agricultural and forest land are therefore converted into shares of total surface area, and the original agricultural and forest land variables are removed.
- Missing numerical values: Remaining missing values are imputed using the median. Median imputation is chosen because several variables have skewed distributions and large outliers.

The same data transformations are applied to the test data so that the training and test datasets have a consistent set of predictors.

3 Prediction

After cleaning and transforming the data, I compare three prediction methods: Elastic Net, Random Forest, and Gradient Boosting. These models range from a regularized linear specification to more flexible tree-based methods that can capture nonlinear relationships and interactions among predictors.

Model performance is evaluated using five-fold cross-validation and Mean Absolute Error (MAE). In five-fold cross-validation, the data are divided into five subsets. The model is trained on four subsets and evaluated on the remaining subset, and this process is repeated so that each subset is used once for validation.

MAE is defined as

$$MAE = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i| ,$$

where y_i is the observed life expectancy and $\hat{y}_i$ is the corresponding prediction. A lower MAE indicates better predictive performance.

Cross-validation is used both to tune model hyperparameters and to compare predictive performance across models before selecting the final model for prediction on the test dataset.

3.1 Elastic Net

Elastic Net is a regularized linear regression method that combines the L_1 penalty of Lasso with the L_2 penalty of Ridge. This makes it useful when predictors may be correlated, as in this dataset, while also allowing less informative variables to be shrunk toward or exactly to zero. Compared with using Ridge or Lasso alone, Elastic Net provides a balance between stabilizing coefficients across correlated predictors and performing variable selection, making it a suitable linear benchmark for prediction in this exercise.

$$\min_{\beta} \left\{ \frac{1}{2n} \sum_{i=1}^{n} (y_i - X_i\beta)^2 + \alpha \left[\rho \sum_{j=1}^{p} |\beta_j| + \frac{1-\rho}{2} \sum_{j=1}^{p} \beta_j^2 \right] \right\}$$

where α controls the overall strength of regularization and ρ determines the relative weight placed on the Lasso (L_1) and Ridge (L_2) penalties.

Elastic Net is included as a prediction model in this exercise for several reasons:

- Provides a linear benchmark
- The Lasso component performs feature selection
- The Ridge component handles correlated predictors
- Regularization reduces overfitting

Because the Elastic Net penalty depends on the scale of the predictors, all predictors are standardized before estimation. The regularization parameters are selected using five-fold cross-validation, with Mean Absolute Error (MAE) used as the model-selection criterion.

3.2 Random Forest

Random Forest is an ensemble learning method that combines predictions from multiple decision trees. Each tree is estimated using a bootstrap sample of the training data, and only a random subset of predictors is considered at each split. Introducing randomness across trees reduces correlation among individual trees and helps improve out-of-sample prediction.

At each node, the model evaluates possible splits among the randomly selected predictors and chooses the split that produces the largest reduction in prediction error, typically measured by the reduction in squared error.

For regression, the final prediction is obtained by averaging predictions across all trees:

$$\hat{y}_i = \frac{1}{B} \sum_{b=1}^{B} \hat{y}_i^{(b)},$$

where B is the number of trees and $\hat{y}_i^{(b)}$ is the prediction from tree b.

Random Forest is included as a prediction model in this exercise for several reasons:

- Captures nonlinear relationships between the predictors and life expectancy
- Relatively robust to outliers
- Allows for interactions among predictors without specifying them explicitly
- Less sensitive to multicollinearity than linear regression
- Averaging across multiple trees helps reduce overfitting

The main hyperparameters considered are the number of trees, the number of predictors considered at each split (max features), maximum tree depth, minimum observations required to split a node (min samples split), and minimum observations in a terminal leaf (min samples leaf). As with Elastic Net, hyperparameters are selected using five-fold cross-validation, with Mean Absolute Error (MAE) used as the model-selection criterion.

3.3 Gradient Boosting

Gradient Boosting is an ensemble learning method that builds decision trees sequentially. Unlike Random Forest, where trees are estimated independently, each new tree in Gradient Boosting is added to improve the errors made by the existing model. In regression, each new tree is fitted to the remaining prediction errors, allowing the model to gradually improve its predictions.

The prediction can be written as

$$\hat{y}_i = \hat{y}_i^{(0)} + \eta \sum_{b=1}^{B} f_b(X_i),$$

where $\hat{y}_i^{(0)}$ is the initial prediction, B is the number of trees, $f_b(X_i)$ is the prediction from tree b, and η is the learning rate that controls the contribution of each new tree.

Gradient Boosting is included as a prediction model in this exercise for several reasons:

- Captures nonlinear relationships between the predictors and life expectancy
- Relatively robust to outliers
- Allows for interactions among predictors without specifying them explicitly
- Less sensitive to multicollinearity
- Controls overfitting through the learning rate and restrictions on tree complexity

The main hyperparameters considered are the number of trees, learning rate, maximum tree depth, minimum observations required to split a node (min samples split) and minimum observations in a terminal leaf (min samples leaf). As with Elastic Net and Random Forest, hyperparameters are selected using five-fold cross-validation, with Mean Absolute Error (MAE) used as the model-selection criterion.

4 Model Results

For each modeling approach, hyperparameters are selected by minimizing the average MAE across five-fold cross-validation. The selected specification within each model class is summarized below.

Table 1: Selected Hyperparameters

Model	Selected Hyperparameters
Elastic Net	$\alpha = 0.1$, $\rho = 0.75$
Random Forest	max depth = 7, max features = 13, trees = 50
	min samples split = 3, min samples leaf = 1
Gradient Boosting	learning rate = 0.1, max depth = 4, trees = 200
	min samples split = 5, min samples leaf = 2

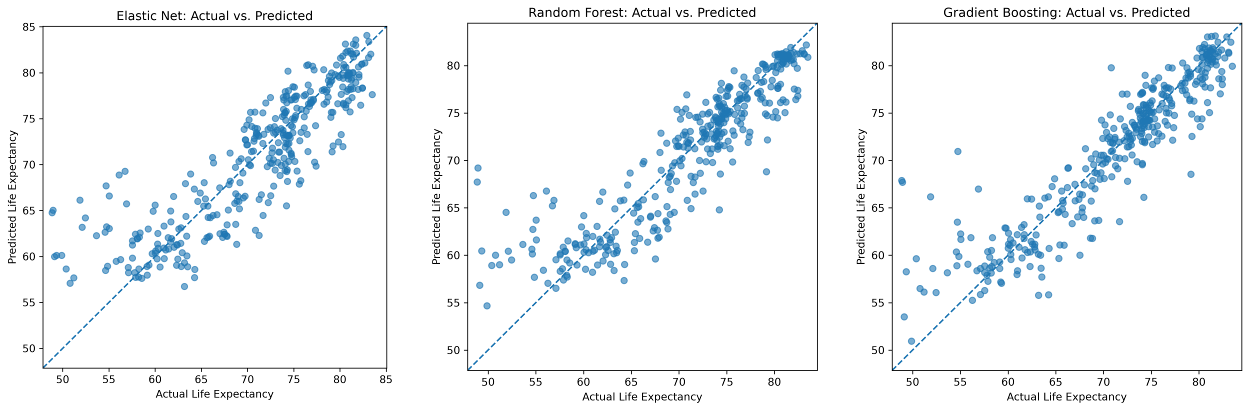
The three selected models are then compared using the same five cross-validation folds. The fold-specific MAEs show how performance varies across validation samples, while the overall comparison is based on the average MAE across the five folds.

Table 2: Five-Fold Cross-Validation Results

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean MAE
Elastic Net	2.955	2.603	3.100	2.931	3.198	2.957
Random Forest	2.446	2.202	2.403	2.231	3.004	2.457
Gradient Boosting	2.280	1.917	2.151	1.925	2.727	2.200

Gradient Boosting achieves the lowest average MAE of approximately 2.20, followed by Random Forest at 2.46 and Elastic Net at 2.96. Gradient Boosting also produces the lowest MAE in each of the five validation folds. Its stronger performance suggests that sequentially correcting errors from earlier trees helps capture useful nonlinear patterns in the data.

The figures below compare actual life expectancy with the out-of-fold predictions from each selected model. The dashed 45-degree line represents perfect prediction, so observations closer to the line indicate smaller prediction errors.

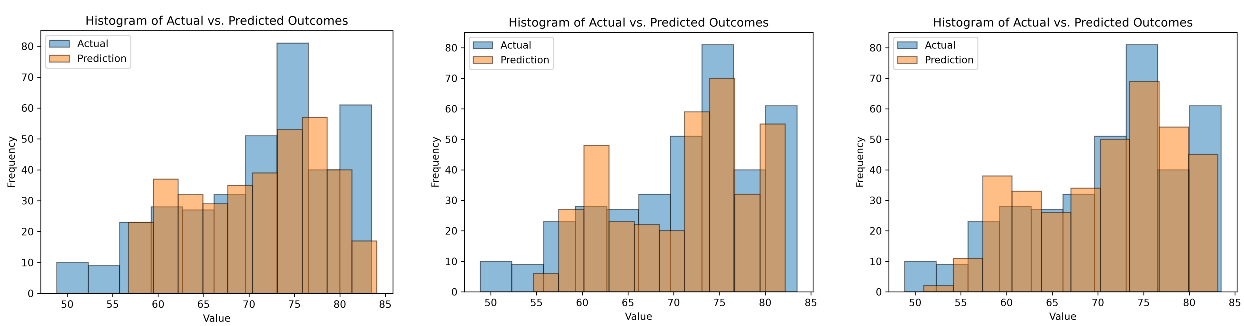


(a) Elastic Net (b) Random Forest (c) Gradient Boosting

Figure 1: Actual versus Predicted Life Expectancy

The figures are consistent with the MAE results. Gradient Boosting predictions are generally more closely concentrated around the 45-degree line, while Random Forest and Elastic Net show larger deviations. All three models tend to predict extreme values closer to the center of the distribution, particularly for observations with low life expectancy.

The histograms below provide an additional comparison of the overall distribution of actual and predicted life expectancy. Greater overlap indicates that the predicted distribution more closely resembles the observed outcome distribution.



(a) Elastic Net (b) Random Forest (c) Gradient Boosting

Figure 2: Distribution of Actual and Predicted Life Expectancy

The histograms reinforce the same pattern. Gradient Boosting more closely reproduces the overall distribution of actual life expectancy. All three models compress predictions toward the center of the distribution, particularly at low levels of life expectancy. This pattern is most pronounced for Elastic Net, while Gradient Boosting captures more of the variation in the observed values.

To better understand the Gradient Boosting model, feature importance is examined for the selected model. Feature importance reflects how much each predictor contributes to reducing prediction error across the trees. It should therefore be interpreted as predictive importance rather than a causal effect. A high importance value does not imply that changes in the predictor cause changes in life expectancy.

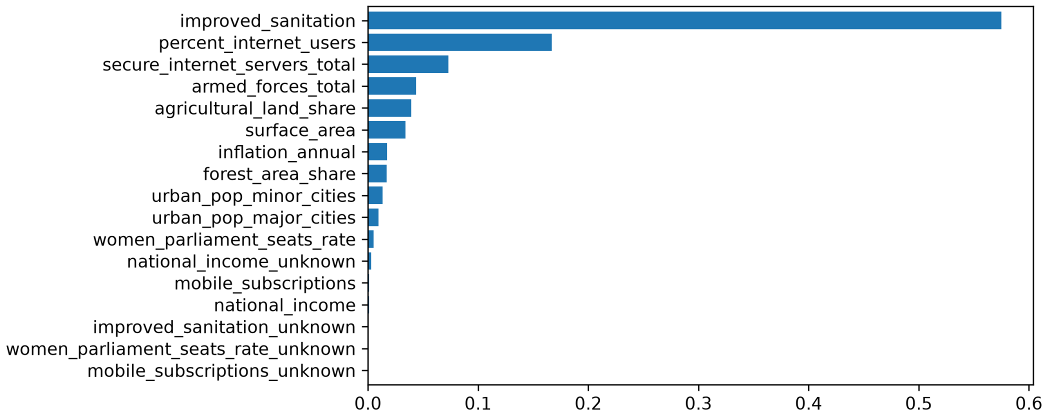


Figure 3: Gradient Boosting Feature Importance

Access to improved sanitation is by far the most important predictor, followed by the share of internet users and the total number of secure internet servers. National income has relatively low feature importance despite a clear increase in average life expectancy across income categories. This likely reflects substantial overlap in predictive information, as national income is highly correlated with internet usage (0.91) and sanitation access (0.79). Once these variables are included in the model, national income contributes relatively little additional predictive information.

To further interpret the Gradient Boosting model, SHAP values are examined. A SHAP value is calculated for each predictor and each observation and measures how much that predictor moves the prediction above or below the model's baseline prediction. Predictors are ranked by their average absolute SHAP value across observations.

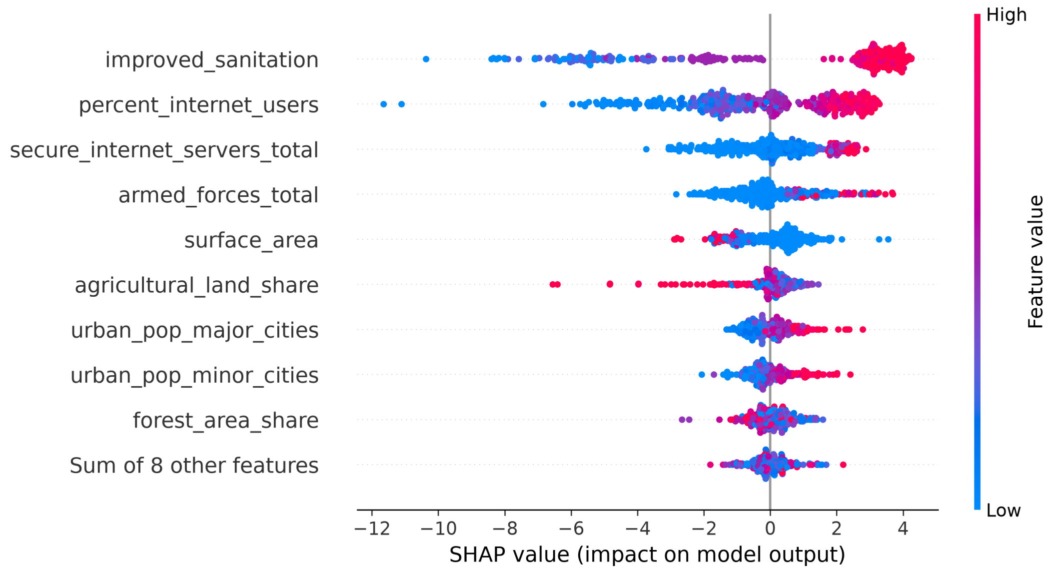


Figure 4: SHAP Values for the Gradient Boosting Model

Each point in the figure represents one observation. Points to the right of zero increase predicted life expectancy, while points to the left decrease it. Red indicates a high value of the predictor and blue indicates a low value.

Improved sanitation has the largest impact on model predictions. High levels of sanitation are associated with higher predicted life expectancy, while low levels substantially reduce the prediction. A similar pattern is observed for internet usage and the number of secure internet servers, where higher values generally increase predicted life expectancy. Higher agricultural land share tends to reduce predicted life expectancy, while higher urban population shares tend to increase it. These relationships describe how the fitted model uses the predictors and should not be interpreted as causal effects.